

# Calculus III Formula Sheet

## 3D Space & Vectors

### Line Equations

**Vector:**  $\vec{r} = \vec{r}_0 + t\vec{v}$

**Parametric:**  $x = x_0 + ta, y = y_0 + tb, z = z_0 + tc$

**Symmetric:**  $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$

### Plane Equations

**Vector:**  $\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$

**Scalar:**  $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

**General:**  $ax + by + cz = d$  where  $d = ax_0 + by_0 + cz_0$

### Quadratic Surfaces

**Ellipsoid:**  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

**Cone:**  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$

**Cylinder:**  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

**Hyperboloid (1 sheet):**  $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

**Hyperboloid (2 sheets):**  $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

**Elliptic Paraboloid:**  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$

**Hyperbolic Paraboloid:**  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}$

### Vector Functions

$\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$

$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$  (tangent vector)

$\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$  (unit tangent)

$\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}$  (unit normal)

$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$  (binormal)

**Arc Length:**  $L = \int_a^b \|\vec{r}'(t)\| dt$

**Curvature:**  $\kappa = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|} = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$

### Partial Derivatives

#### Notation

$f_x = \frac{\partial f}{\partial x}, f_y = \frac{\partial f}{\partial y}, f_z = \frac{\partial f}{\partial z}$

$f_{xx} = \frac{\partial^2 f}{\partial x^2}, f_{xy} = \frac{\partial^2 f}{\partial y \partial x}, f_{yy} = \frac{\partial^2 f}{\partial y^2}$

**Clairaut's Theorem:**  $f_{xy} = f_{yx}$  (if continuous)

#### Differentials

$df = f_x dx + f_y dy + f_z dz$

#### Chain Rule

If  $z = f(x, y)$ ,  $x = g(t)$ ,  $y = h(t)$ :

$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$

If  $z = f(x, y)$ ,  $x = g(s, t)$ ,  $y = h(s, t)$ :

$\frac{\partial z}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s}$

$\frac{\partial z}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}$

**General:**  $\frac{\partial z}{\partial t_i} = \sum_{j=1}^n \frac{\partial z}{\partial x_j} \frac{\partial x_j}{\partial t_i}$

#### Implicit Differentiation

If  $F(x, y) = 0$ :  $\frac{dy}{dx} = -\frac{F_x}{F_y}$

If  $F(x, y, z) = 0$ :  $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$

### Directional Derivatives & Gradient

**Gradient:**  $\nabla f = \langle f_x, f_y, f_z \rangle$

**Directional Derivative:**  $D_{\vec{u}} f = \nabla f \cdot \vec{u}$

Max rate of change:  $\|\nabla f\|$  in direction of  $\nabla f$

$\nabla f$  is orthogonal to level curves/surfaces

### Tangent Planes & Linear Approximation

**Tangent Plane to  $z = f(x, y)$  at  $(x_0, y_0, z_0)$ :**

$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

**Tangent Plane to  $F(x, y, z) = k$  at  $(x_0, y_0, z_0)$ :**

$F_x(x - x_0) + F_y(y - y_0) + F_z(z - z_0) = 0$

**Normal Line:**  $\vec{r}(t) = \langle x_0, y_0, z_0 \rangle + t\nabla F(x_0, y_0, z_0)$

## Optimization

### Critical Points

$\nabla f = \vec{0}$  or  $\nabla f$  does not exist

### Second Derivative Test

**Hessian:**  $D = f_{xx}f_{yy} - (f_{xy})^2$  at  $(a, b)$

•  $D > 0$  and  $f_{xx} > 0$ : Relative minimum

•  $D > 0$  and  $f_{xx} < 0$ : Relative maximum

•  $D < 0$ : Saddle point

•  $D = 0$ : Inconclusive

### Lagrange Multipliers

To optimize  $f(x, y, z)$  subject to  $g(x, y, z) = k$ :

Solve  $\nabla f = \lambda \nabla g$  and  $g(x, y, z) = k$

**Multiple Constraints:**  $g_1 = k_1, g_2 = k_2$ :

$\nabla f = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$

### Double Integrals

#### Rectangular Regions

$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy$

If  $f(x, y) = g(x)h(y)$ :

$\iint_R g(x)h(y) dA = \left( \int_a^b g(x) dx \right) \left( \int_c^d h(y) dy \right)$

#### General Regions

**Type I:**  $D = \{(x, y) | a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$

$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$

**Type II:**  $D = \{(x, y) | c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$

$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$

#### Polar Coordinates

$x = r \cos \theta, y = r \sin \theta, dA = r dr d\theta$

$\iint_D f(x, y) dA = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$

**Area:**  $A = \iint_D dA$

#### Triple Integrals

##### Rectangular Box

$\iiint_B f(x, y, z) dV = \int_r^s \int_c^d \int_a^b f(x, y, z) dx dy dz$

##### General Regions

**Type I:**  $(x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)$

$\iiint_E f dV = \iint_D \left[ \int_{u_1}^{u_2} f(x, y, z) dz \right] dA$

**Type II:**  $(y, z) \in D, u_1(y, z) \leq x \leq u_2(y, z)$

**Type III:**  $(x, z) \in D, u_1(x, z) \leq y \leq u_2(x, z)$

#### Cylindrical Coordinates

$x = r \cos \theta, y = r \sin \theta, z = z$

$dV = r dz dr d\theta$

$\iiint_E f dV = \int_\alpha^\beta \int_{h_1}^{h_2} \int_{u_1}^{u_2} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$

#### Spherical Coordinates

$x = \rho \sin \varphi \cos \theta, y = \rho \sin \varphi \sin \theta, z = \rho \cos \varphi$

$x^2 + y^2 + z^2 = \rho^2$

$dV = \rho^2 \sin \varphi d\rho d\theta d\varphi$

$0 \leq \theta \leq 2\pi, 0 \leq \varphi \leq \pi$

#### Change of Variables (Jacobian)

**2D:**  $dA = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$  where

$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$

**3D:**  $dV = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw$  where

$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$

#### Surface Area

For  $z = f(x, y)$  over region  $D$ :

$S = \iint_D \sqrt{1 + \left( \frac{\partial f}{\partial x} \right)^2 + \left( \frac{\partial f}{\partial y} \right)^2} dA$

For parametric surface  $\vec{r}(u, v)$  over  $D$ :

$S = \iint_D \|\vec{r}_u \times \vec{r}_v\| dA$

**Vector Fields**

$\vec{F}(x,y,z) = P\hat{i} + Q\hat{j} + R\hat{k}$

$\vec{F}(x,y) = P\hat{i} + Q\hat{j}$

**Gradient Field:**  $\vec{F} = \nabla f$

**Conservative:**  $\vec{F} = \nabla f$  for some potential  $f$

**Line Integrals**

**With Respect to Arc Length**

$\int_C f(x,y,z) \, ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| \, dt$

**Direction independent**

**With Respect to Coordinates**

$\int_C f(x,y) \, dx = \int_a^b f(x(t),y(t))x'(t) \, dt$

$\int_C f(x,y) \, dy = \int_a^b f(x(t),y(t))y'(t) \, dt$

**Combined:**  $\int_C P \, dx + Q \, dy + R \, dz$

**Direction dependent:**  $\int_{-C} f \, dx = - \int_C f \, dx$

**Of Vector Fields**

$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt$

$= \int_C P \, dx + Q \, dy + R \, dz$

$= \int_C \vec{F} \cdot \vec{T} \, ds$

**Work Interpretation:** Work done by force field  $\vec{F}$  along  $C$

**Conservative Vector Fields**

**Test ( $\mathbb{R}^2$ ):**  $\vec{F} = P\hat{i} + Q\hat{j}$  is conservative if

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$  (on simply connected region)

**Test ( $\mathbb{R}^3$ ):**  $\vec{F}$  is conservative if  $\text{curl } \vec{F} = \vec{0}$

**Finding Potential:** Solve  $\frac{\partial f}{\partial x} = P, \frac{\partial f}{\partial y} = Q, \frac{\partial f}{\partial z} = R$

**Fundamental Theorem for Line Integrals**

If  $\vec{F} = \nabla f$ :  $\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$

Path independent:  $\int_C \vec{F} \cdot d\vec{r} = 0$  for closed  $C$

**Green's Theorem**

$C$  positively oriented, simple closed curve,  $D$  region inside:

$\oint_C P \, dx + Q \, dy = \iint_D \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$

**Vector Form:**  $\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\text{curl } \vec{F}) \cdot \hat{k} \, dA$

**Flux Form:**  $\oint_C \vec{F} \cdot \vec{n} \, ds = \iint_D (\text{div } \vec{F}) \, dA$

**Area:**  $A = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C x \, dy - y \, dx$

**Curl & Divergence**

**Curl**

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$= \left( \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} + \left( \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \hat{j} + \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k}$$

**Properties:**

$\text{curl}(\nabla f) = \vec{0}$

If  $\text{curl } \vec{F} = \vec{0}$  on  $\mathbb{R}^3$ , then  $\vec{F}$  is conservative

**Divergence**

$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$

**Properties:**

$\text{div}(\text{curl } \vec{F}) = 0$

$\nabla^2 f = \text{div}(\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$  (Laplacian)

**Surface Integrals**

**Parametric Surfaces**

$\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$

**Tangent Vectors:**

$\vec{r}_u = \frac{\partial x}{\partial u} \hat{i} + \frac{\partial y}{\partial u} \hat{j} + \frac{\partial z}{\partial u} \hat{k}$

$\vec{r}_v = \frac{\partial x}{\partial v} \hat{i} + \frac{\partial y}{\partial v} \hat{j} + \frac{\partial z}{\partial v} \hat{k}$

**Normal Vector:**  $\vec{n} = \vec{r}_u \times \vec{r}_v$

**Surface Integral (Scalar)**

For  $z = g(x,y)$  over  $D$ :

$\iint_S f(x,y,z) \, dS = \iint_D f(x,y,g(x,y)) \sqrt{1 + g_x^2 + g_y^2} \, dA$

For parametric  $\vec{r}(u,v)$  over  $D$ :

$\iint_S f(x,y,z) \, dS = \iint_D f(\vec{r}(u,v)) \|\vec{r}_u \times \vec{r}_v\| \, dA$

**Surface Integral (Vector Field / Flux)**

$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n} \, dS$

For  $z = g(x,y)$  (upward orientation):

$\iint_S \vec{F} \cdot d\vec{S} = \iint_D (-Pg_x - Qg_y + R) \, dA$

For parametric  $\vec{r}(u,v)$ :

$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F}(\vec{r}(u,v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA$

**Stokes' Theorem**

$S$  oriented surface,  $C$  boundary with positive orientation:

$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot d\vec{S}$

**Divergence Theorem**

$E$  solid region,  $S$  boundary with outward orientation:

$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \text{div } \vec{F} \, dV$

**Common Parametrizations**

**Sphere** ( $x^2 + y^2 + z^2 = a^2$ ):

$\vec{r}(\theta, \varphi) = a \sin \varphi \cos \theta \hat{i} + a \sin \varphi \sin \theta \hat{j} + a \cos \varphi \hat{k}$

$0 \leq \theta \leq 2\pi, \quad 0 \leq \varphi \leq \pi$

**Cylinder** ( $x^2 + y^2 = a^2$ ):

$\vec{r}(z, \theta) = a \cos \theta \hat{i} + a \sin \theta \hat{j} + z \hat{k}$

$0 \leq \theta \leq 2\pi$

**Cone** ( $z = \sqrt{x^2 + y^2}$ ):

$\vec{r}(r, \theta) = r \cos \theta \hat{i} + r \sin \theta \hat{j} + r \hat{k}$

$0 \leq \theta \leq 2\pi$